

Plant Biotechnology

Lab. BT 421

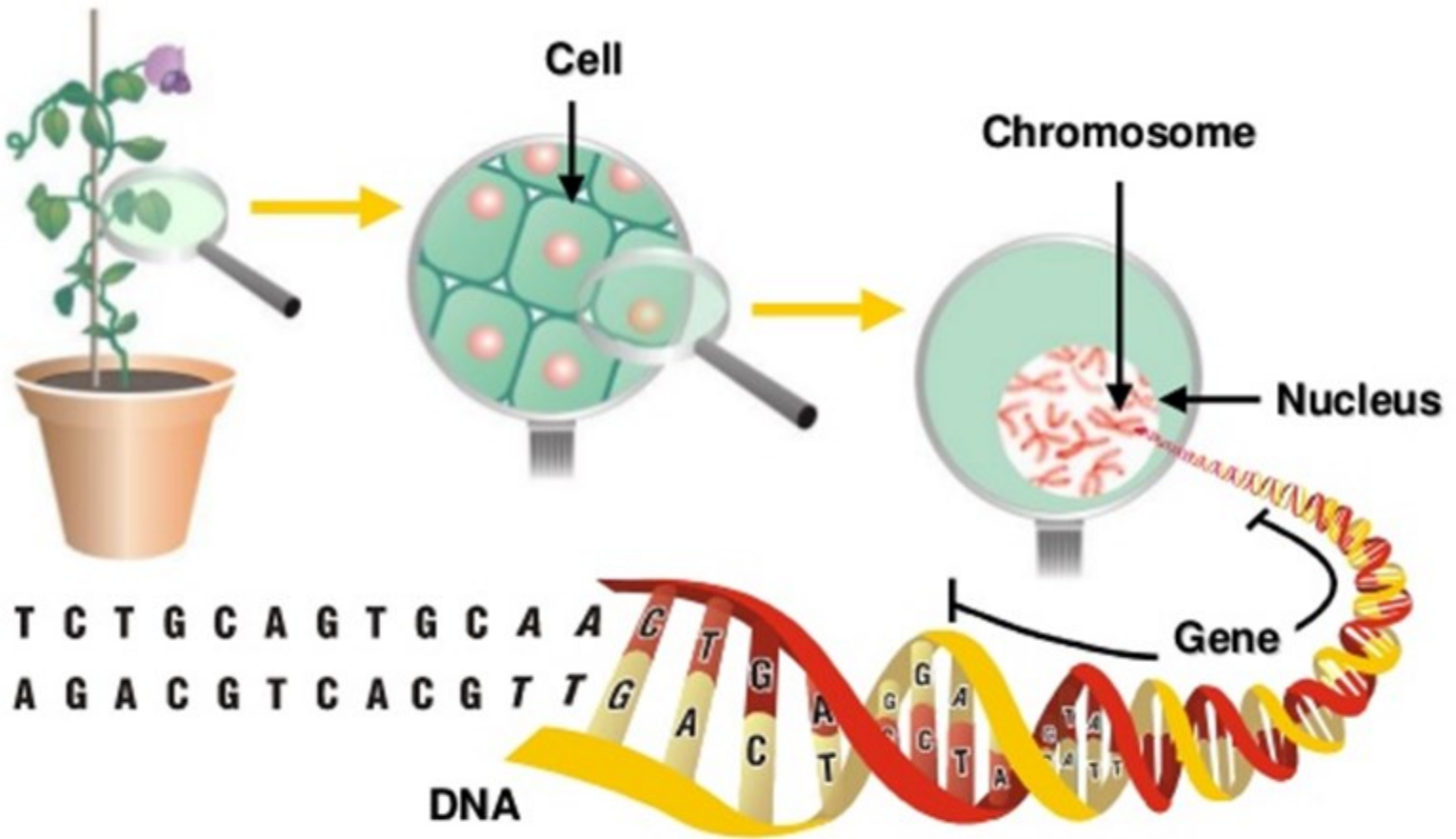
Lab (9)

**DNA extraction from
plant tissues**

By:

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DNA in Plants



General DNA extraction protocol

- ❖ **DNA can be isolated from a number of plant tissues. Young leaf tissue is a good source of DNA.**
- ❖ **Extraction procedures for plant DNA in general must accomplish the following:**
 1. **The cell walls must be broken (or digested away) in order to release the cellular constituents. This is usually done by grinding the tissue in dry ice or liquid nitrogen with a mortar and pestle or a food grinder.**

General DNA extraction protocol

- 2. The cell membranes must be disrupted, so that the DNA is released into the extraction buffer.**

This is accomplished by using a detergent, usually **SDS (sodium dodecyl sulfate).**

General DNA extraction protocol

3. The DNA must be protected from the endogenous nucleases. The detergents are used for this purpose, as is EDTA (ethylenediaminetetraacetic acid).

EDTA is a chelating agent that binds magnesium ions, generally considered a necessary cofactor for most nucleases. In addition, the buffer/tissue mixture is emulsified with either chloroform or phenol to denature and separate the proteins from the DNA.

General DNA extraction protocol

- 4. Shearing of the DNA should be minimized. DNA in solution can be broken by exposure to turbulence (e.g., being quickly drawn through a small orifice).**
- 5. The time between thawing of the frozen, pulverized tissue and its exposure to the extraction buffer should be minimized to avoid nucleolytic degradation of the DNA.**

DNA Extraction procedure (Spinach) Phenol-Chloroform

- 1. Weight out 20 grams of plant tissue (Spinach).**
- 2. Surface sterilization (as in the protoplast isolation protocol).**
- 3. Fragmentation: chop the tissue into small pieces using a sterile single edge razor blade.**
- 4. Add 20 ml of lysis buffer to the spinach.**



A blade

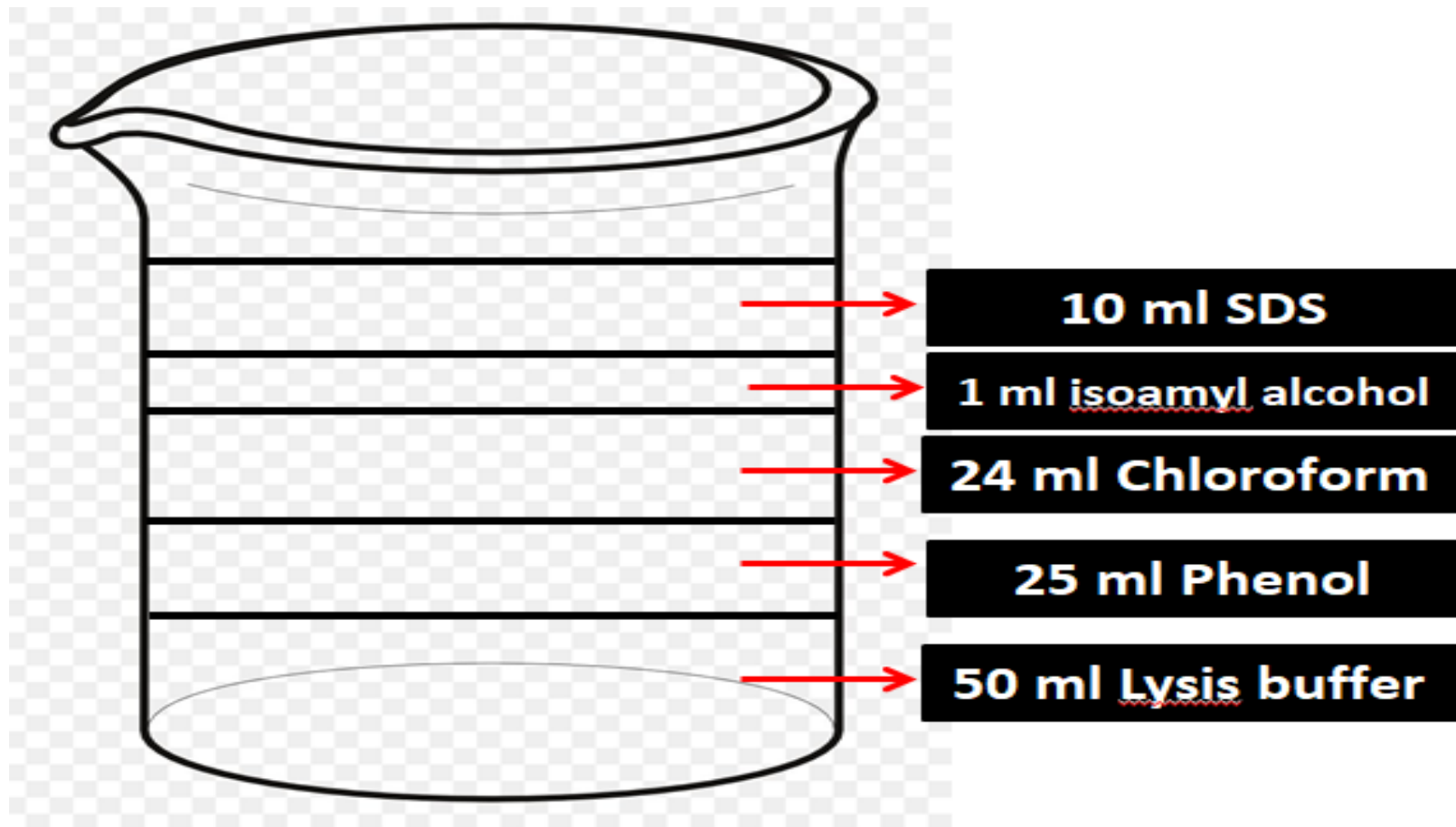
DNA Extraction procedure

5. **Grind** the spinach tissue gently using pistle and mortar to break the cell wall and cell membrane and releasing the DNA by the help of lysis buffer (pH=8).
6. Immediately transfer the mixture to a **cleaned cheese cloth** to filter it into a beaker.
7. Take 50 ml of the **lysis extract** using a cylinder and pour it into another cleaned 500 ml beaker.



DNA Extraction procedure

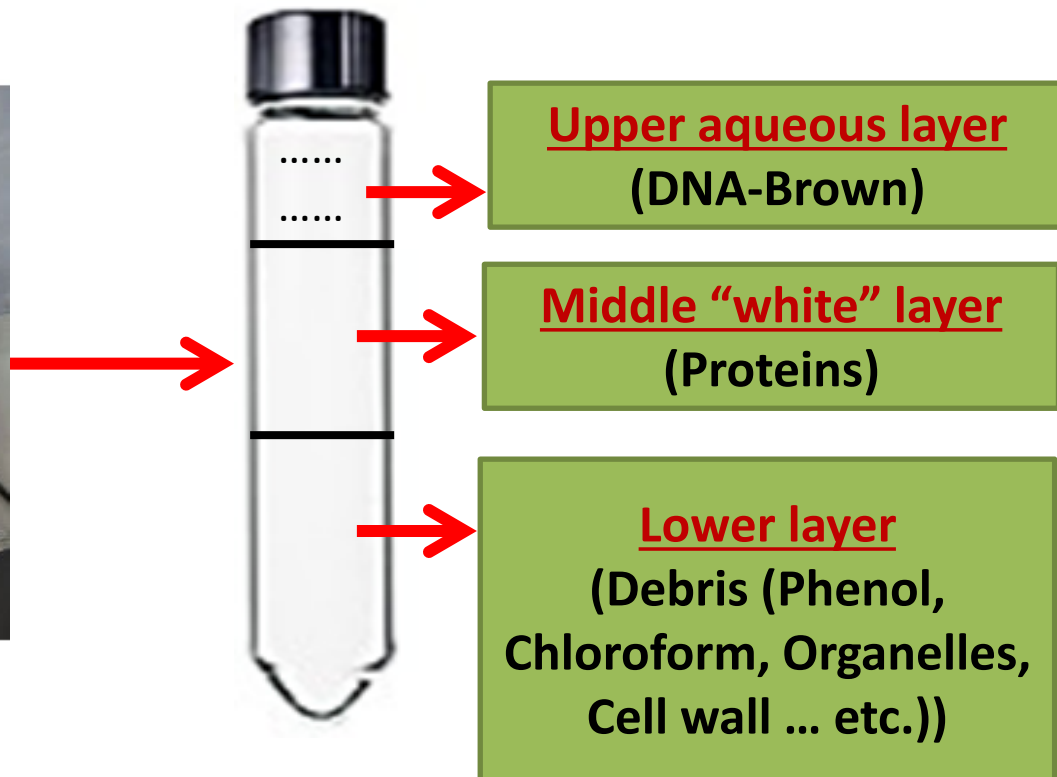
8. Add the following chemical solutions into the lysis extract and shake well by hand for 10 minutes.



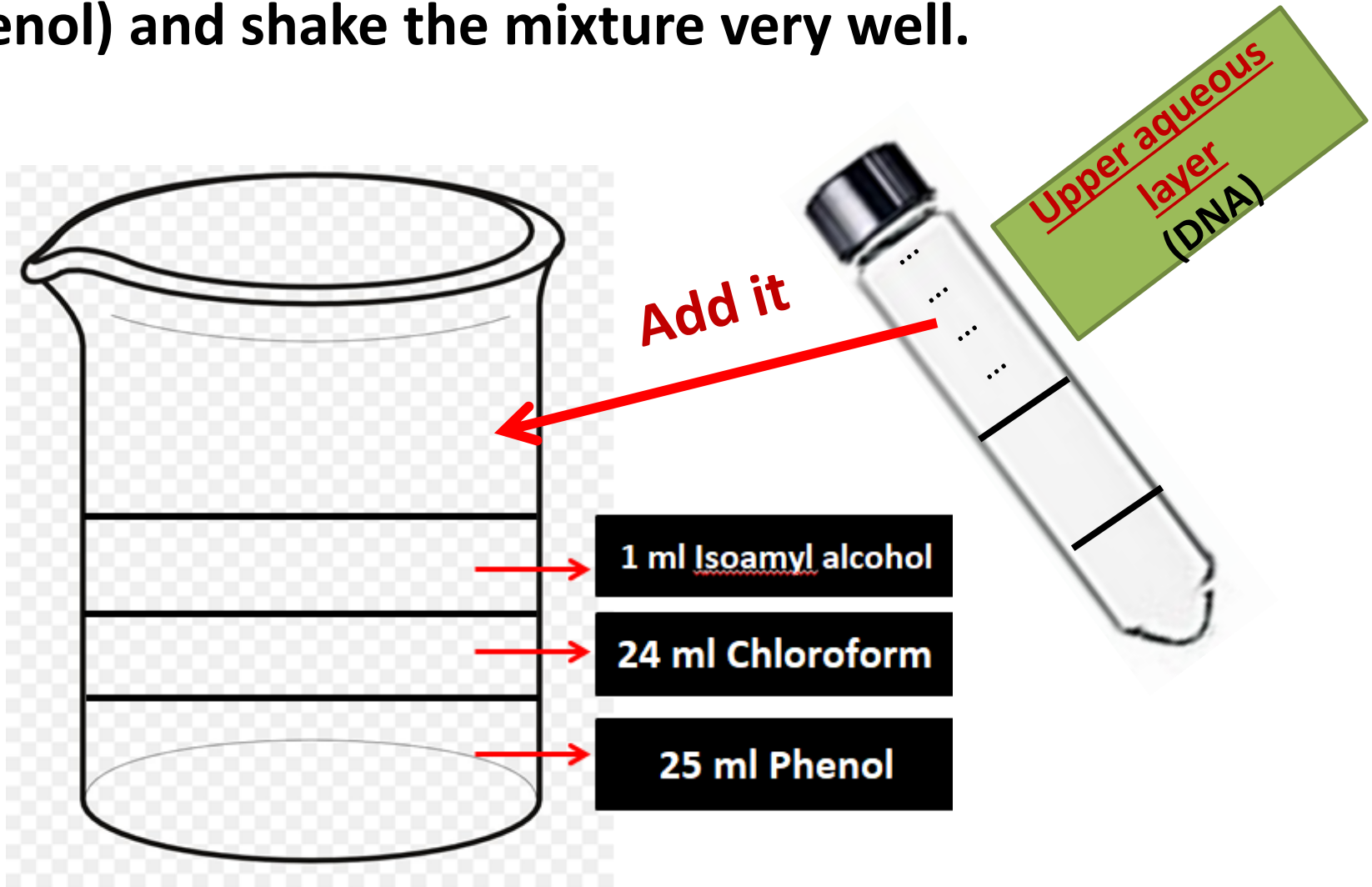
DNA Extraction procedure (from Spinach)

9. **Centrifuge** at 4000 rpm for 15 min.

If possible, use a refrigerated centrifuge tubes set to 4 C).

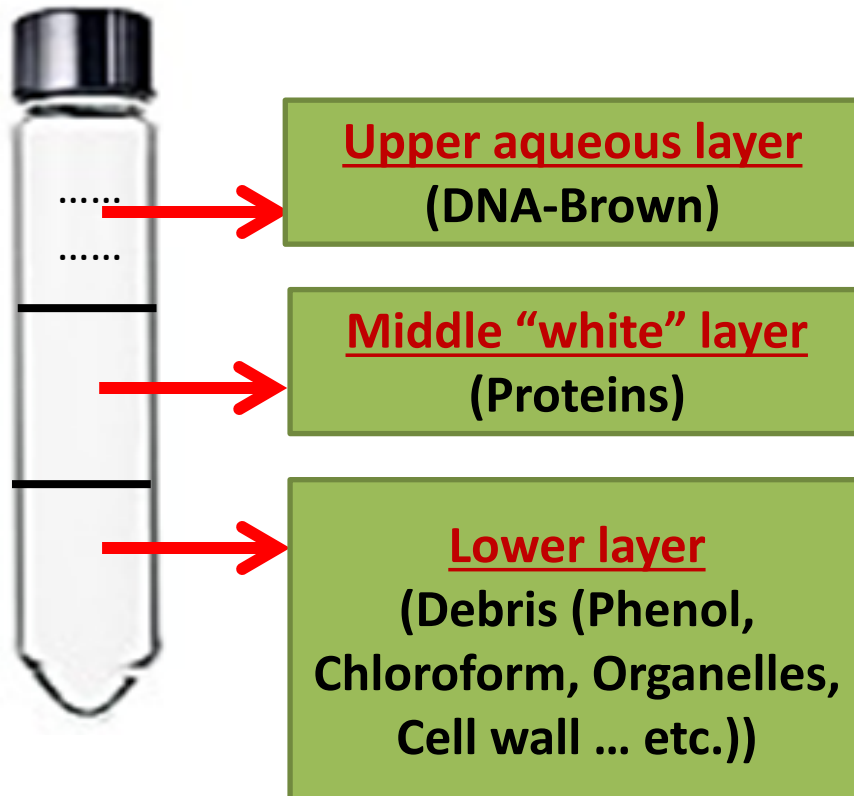


10. Pipette out the **aqueous layer (DNA)** carefully then add it into a previously prepared mixture in a beaker composed of (**1 ml** Isoamyl alcohol, **24 ml** chloroform and **25 ml** phenol) and shake the mixture very well.



DNA Extraction procedure

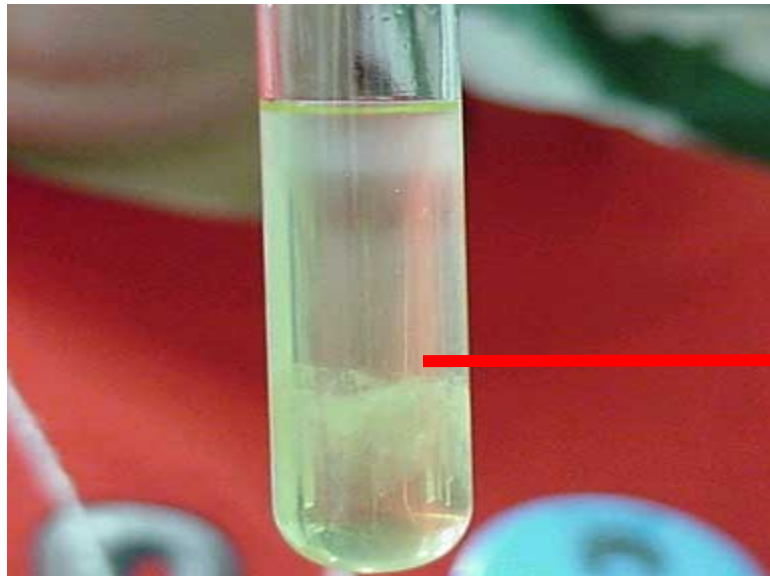
11. **Centrifuge** at 4000 rpm for 15 min.
(Second Purification).



DNA Extraction procedure

12. Pipette out the **aqueous layer (DNA)** carefully then precipitate DNA by adding 2 volumes of **95% cold ethanol**.

13. Add 20:1 **3M Potassium acetate** into DNA and swirl with a glass rod.



Fibrous
chromosomal DNA
Indicated by
Bubbles + Turbidity

DNA Extraction procedure

14. Keep DNA in the refrigerator for the next lab.

Note: **Phenol and chloroform** are **toxic** chemicals so **DON'T** smell them directly or touch them without wearing gloves.



These are 3 helpful links to understand the **Plant DNA Extraction protocol** more, please watch them.

PLANT GENOMIC DNA ISOLATION:

<https://youtu.be/9-olhcH4NI4>

DNA extraction from plants:

<https://youtu.be/PgwWuTDaPd4>

DNA Isolation from Neem Leaf:

<https://youtu.be/yooQYMAZvJw>